

Complete Scenario Design, RoadRunner 3D & MATLAB Simulation Guide

A Flawless Step-by-Step Engineering Manual for Designing Indian Road Environments (Village Roads, Unregulated Intersections, Potholes, Cattle Crossing) in MATLAB Driving Scenario Designer and RoadRunner 3D.

1. Method 1: Driving Scenario Designer (Visual Drag-and-Drop GUI)

The **Driving Scenario Designer** app in MATLAB enables rapid, code-free visual creation of roads, actors, and trajectories:

Step #	Action Item	Detailed Instructions & Parameters
Step 1	Launch App	In MATLAB Command Window, type: <code>drivingScenarioDesigner</code> and press Enter. A visual 3D canvas window will appear.
Step 2	Add Road	Click ' Add Road ' in the top toolbar. Click on the canvas at (0,0), (50,0), and double-click at (100,0) to create a straight track. In the right panel, set Road Width = 7.0m and Lanes = 2 .
Step 3	Add Ego Car	Click ' Add Ego Vehicle '. Click on the start of the road at (0,0). Set its initial speed to 30 km/h (8.3 m/s) .
Step 4	Add Actors (Cow/Auto/People)	Click ' Add Actor '. Select <i>Car, Truck, Pedestrian, or Custom Actor (Cattle)</i> . Click on the roadside to place it, then click waypoints to define its moving trajectory (e.g. crossing right-to-left).
Step 5	Mount Sensors	Select the Ego vehicle, click ' Add Sensor ' → choose Camera (FOV: 100°) and LiDAR (360° Scanner, 80m Range) .
Step 6	Simulate & Export	Click the green ' Run ' button to watch live 3D animation. Then click Export → Export MATLAB Function to auto-generate a clean .m script!

2. Method 2: RoadRunner 3D Photorealistic Map Design

RoadRunner is MathWorks' interactive editor for building realistic 3D road scenes and exporting OpenDRIVE (.xodr) networks:

RoadRunner Step	Specific Action & Tool	Indian Road Customization Settings
1. Road Plan Tool	Press Shortcut ' R '	Draw a 2-lane road with curves. Set total road width to 7.0 meters.
2. Lane Marking Removal	Lane Marking Tool	Select center/edge markings and set them to ' None ' to simulate unstructured, unmarked Indian asphalt.
3. Surface Damage & Potholes	Decal Tool	From the Asset Library, drag ' Pothole / Road Crack ' decals directly onto the asphalt surface at 25m and 60m.
4. Foliage & Props	Prop Placement Tool	Drag Indian roadside trees, wooden fences, concrete barriers, and shop buildings along the road perimeter.
5. OpenDRIVE Export	File → Export → OpenDRIVE	Export as <code>indian_village_road.xodr</code> for direct import into MATLAB Driving Scenario.

3. 5 Mandatory Indian Test Scenarios Specifications

Scenario ID & Name	Road Characteristics	Actors & Speed Parameters	Avoidance / Control Expectation
1. Unmarked Village Road	Width: 7.0m, No lane lines, dirt shoulders, 2 potholes.	- Oncoming Tractor ($v = -4.5$ m/s) - Pedestrian on shoulder ($v = 0$)	Swerves around potholes while maintaining clearance from oncoming tractor.
2. 4-Way Urban Intersection	Width: 8.0m, Unregulated 4-way cross without traffic signals.	- Crossing Auto-Rickshaw ($v = +3.2$ m/s) - Weaving 2-Wheeler ($v = -3.0$ m/s)	Yields to crossing auto-rickshaw, then accelerates through open gap.

3. Highway Merge	Width: 9.0m, 3-Lane road with high-speed traffic.	- Leading slow auto ($v = 5.0$ m/s) - Merging heavy truck ($v = 7.5$ m/s)	Executes smooth Frenet overtake maneuver into middle/outer lane.
4. Dense Market Area	Width: 7.0m, High roadside activity & pushcarts.	- Jaywalking pedestrian ($v = +0.8$ m/s) - Auto-rickshaw ($v = 2.5$ m/s)	Reduces nominal speed to 18 km/h, buffers around pedestrians.
5. Sudden Cattle Crossing	Width: 7.0m, Straight road with low Time-to-Collision.	- Cow steps in at $t \geq 1.5$s ($v = -1.1$ m/s) - Oncoming bike ($v = -4.0$ m/s)	Executes rapid risk evaluation & smooth left-corridor avoidance curve.

4. Sensor Mounting Architecture & Real-Time Data Extraction

Sensor Type	Mounting Coordinates (X, Y, Z)	Field of View & Range Specs	Output Data Extracted
■ Front Camera (Vision)	[1.9m, 0.0m, 1.4m] Mounted behind windshield center	- Horizontal FOV: 100° - Vertical FOV: 20° - Max Range: 60.0 meters	2D Bounding Boxes, Object Class IDs (Auto, Cow, Pothole), Range & Bearing
■ 3D LiDAR (Laser Scanner)	[1.0m, 0.0m, 1.8m] Mounted on top roof center	- Azimuth: [-180°, +180°] (360°) - Elevation: [-15°, +15°] - Max Range: 80.0 meters	3D Point Clouds (X, Y, Z, Intensity), 3D obstacle clusters, exact depth

MATLAB Sensor Mounting & Data Extraction Code:

```
% 1. Mount Front Camera & Roof LiDAR on Ego Vehicle
cameraSensor = visionDetectionGenerator('SensorIndex', 1, 'SensorLocation', [1.9 0 1.4], ...
'MaxRange', 60, 'FieldOfView', [100 20], 'DetectionProbability', 0.95);

lidarSensor = lidarPointCloudGenerator('SensorIndex', 2, 'SensorLocation', [1.0 0 1.8], ...
'MaxRange', 80, 'AzimuthLimits', [-180 180], 'ElevationLimits', [-15 15]);

% 2. Extract Data in Closed-Loop Simulation Loop
while advance(scenario)
    time = scenario.SimulationTime;
    actors = targetPoses(egoVehicle); % Surrounding traffic & obstacles
    [cameraDets, numVision] = cameraSensor(actors, time); % 2D Detections & Class IDs
    [ptCloud, numPts] = lidarSensor(actors, time); % 3D Point Cloud (X, Y, Z)
    fusedTracks = sensorFusion(cameraDets, ptCloud); % Pass to Frenet Planner
end
```

5. How to Load and Execute Scenarios in MATLAB

```
% =====
% MATLAB COMMAND WINDOW: RUNNING ANY SCENARIO IN 1 CLICK
% =====
% 1. Set current directory in MATLAB
cd('d:/SIH2026/matlab_pipeline');

% 2. Run Scenario 5 (Sudden Cattle Crossing) with Live Bird's Eye View
main_closed_loop_sim(5);

% 3. Run Scenario 1 (Unmarked Village Road with Potholes)
main_closed_loop_sim(1);

% 4. Load RoadRunner OpenDRIVE Scene Programmatically
scenario = drivingScenario('SampleTime', 0.1);
roadNetwork(scenario, 'OpenDRIVE', 'd:/SIH2026/matlab_pipeline/roadrunner/indian_village.xodr');
plot(scenario);
```

6. Quality Verification & Submission Checklist

- **Collision-Free Guarantee:** Verify in all 5 scenarios that Total Collisions = 0 and Minimum Clearance is at least 1.5 meters from obstacles.
- **Camera-LiDAR Sensor Coverage:** Ensure Camera FOV is 100° to 120° and LiDAR scanner covers full 360° up to 80m range.
- **Simulation Sampling Rate:** Keep SampleTime = 0.1s (10 Hz) for realistic automotive control cycles.
- **Video Demonstration Recording:** Use MATLAB VideoWriter or OBS screen recording to capture the 3D Bird's Eye View simulation during obstacle avoidance.